

1 Problem 4

Part a)

Let $(X_k)_{k \in \mathbb{N}}$ be a sequence of i.i.d. Bernoulli(p) random variables indicating whether the k -th ticket holder shows up. For N sold tickets, the number of arriving passengers is $S_N = \sum_{k=1}^N X_k$. Of these, $\min(S_N, S)$ board and any remainder are surplus. Earning a per boarding passenger and losing b per surplus, the profit is:

$$G_N = a \min(S_N, S) - b(S_N - S) \cdot \mathbf{1}_{\{S_N > S\}}.$$

Notice that $S_{N+1} = S_N + X_{N+1}$. Since S_N is a function of $\{X_1, \dots, X_N\}$ and the sequence $(X_k)_{k \in \mathbb{N}}$ is mutually independent (Definition 3.2.5), X_{N+1} is independent of S_N . We analyse the relevant cases of the value of X_{N+1} and the position of S_N relative to S .

- $X_{N+1} = 0$: The $(N+1)$ -th ticket holder does not show up, so $S_{N+1} = S_N$ and thus $G_{N+1} = G_N$, giving:

$$G_{N+1} - G_N = 0.$$

- $X_{N+1} = 1$ **and** $S_N < S$: Since $S_N \leq S-1$, The $(N+1)$ -th ticket holder boards the plane. Therefore $\min(S_{N+1}, S) = S_N + 1$ and the penalty term is zero in both G_{N+1} and G_N . Thus

$$G_{N+1} - G_N = a(S_N + 1) - aS_N = a.$$

- $X_{N+1} = 1$ **and** $S_N \geq S$: The $(N+1)$ -th ticket holder does not board the plane. The revenue part is unchanged since $\min(S_{N+1}, S) = S = \min(S_N, S)$, but the airline has a loss of b . Thus

$$G_{N+1} - G_N = [aS - b(S_N + 1 - S)] - [aS - b(S_N - S)] = -b.$$

Combining all cases using indicator functions, we obtain

$$G_{N+1} - G_N = X_{N+1} \cdot [a \mathbf{1}_{\{S_N < S\}} - b \mathbf{1}_{\{S_N \geq S\}}].$$

Since X_{N+1} is independent of S_N , it is also independent of the function $a \mathbf{1}_{\{S_N < S\}} - b \mathbf{1}_{\{S_N \geq S\}}$ of S_N . Because bounded random variables are in $L^1(\Omega, \mathcal{F}, \mathbb{P})$, we can apply Theorem 3.3.32 (expectation of independent products) and Lemma 3.3.9 (linearity):

$$\begin{aligned} \mathbb{E}[G_{N+1} - G_N] &= \mathbb{E}\left[X_{N+1} (a \mathbf{1}_{\{S_N < S\}} - b \mathbf{1}_{\{S_N \geq S\}})\right] \\ &= \mathbb{E}[X_{N+1}] \cdot (a \mathbb{E}[\mathbf{1}_{\{S_N < S\}}] - b \mathbb{E}[\mathbf{1}_{\{S_N \geq S\}}]) \\ &= p \cdot (a \mathbb{P}(S_N < S) - b \mathbb{P}(S_N \geq S)) \\ &= p \cdot [a \mathbb{P}(S_N < S) - b(1 - \mathbb{P}(S_N < S))] \\ &= p \cdot [(a+b) \mathbb{P}(S_N < S) - b]. \end{aligned}$$

Since $p > 0$, we have $\mathbb{E}[G_{N+1}] - \mathbb{E}[G_N] > 0$ if and only if $(a+b) \mathbb{P}(S_N < S) - b > 0$. Rearranging:

$$\begin{aligned} \mathbb{E}[G_{N+1}] - \mathbb{E}[G_N] > 0 &\iff (a+b) \mathbb{P}(S_N < S) > b \\ \mathbb{E}[G_{N+1}] > \mathbb{E}[G_N] &\iff \mathbb{P}(S_N < S) > \frac{b}{a+b}, \end{aligned}$$

□

Part b)

Step 1: Distribution of S_N and CLT approximation. Since each of the N ticket holders shows up independently with probability $p = 0.95$, and $S_N = \sum_{k=1}^N X_k$ with $(X_k)_{k \in \mathbb{N}}$ i.i.d. Bernoulli(p), we have

$$S_N \sim \text{Bin}(N, p), \quad \mathbb{E}[S_N] = Np, \quad \text{Var}(S_N) = Np(1-p).$$

Let $q := 1 - p = 0.05$. By the de Moivre-Laplace theorem (Corollary 4.4.3), for all $-\infty \leq a < b \leq \infty$,

$$\mathbb{P}\left(a \leq \frac{S_N - Np}{\sqrt{Npq}} \leq b\right) \xrightarrow{N \rightarrow \infty} \Phi(b) - \Phi(a).$$

Because S_N takes integer values, we apply the continuity correction from Remark 4.4.4. Setting $a = -\infty$ and $k = S - 1$, the fine approximation yields:

$$\begin{aligned} \mathbb{P}(S_N < S) &= \mathbb{P}(-\infty < S_N \leq S - 1) \\ &= \mathbb{P}\left(-\infty < S_N \leq S - \frac{1}{2}\right) \\ &= \mathbb{P}\left(-\infty < \frac{S_N - Np}{\sqrt{Npq}} \leq \frac{S - \frac{1}{2} - Np}{\sqrt{Npq}}\right) \\ &\stackrel{4.4.4}{\approx} \Phi\left(\frac{S - \frac{1}{2} - Np}{\sqrt{Npq}}\right) - \Phi(-\infty) \\ &= \Phi\left(\frac{S - \frac{1}{2} - Np}{\sqrt{Npq}}\right). \end{aligned}$$

Step 2: Monotonicity and the optimal overbooking level. Since $X_{N+1} \geq 0$ P-a.s., we have $S_{N+1} = S_N + X_{N+1} \geq S_N$, so:

$$\begin{aligned} \{S_{N+1} < S\} &\subseteq \{S_N < S\} \\ \implies \mathbb{P}(S_{N+1} < S) &\leq \mathbb{P}(S_N < S). \end{aligned}$$

The map $N \mapsto \mathbb{P}(S_N < S)$ is therefore non-increasing. By part a), the $(N + 1)$ -th ticket yields a positive expected contribution if and only if $\mathbb{P}(S_N < S) > b/(a + b)$. Since this condition becomes harder to satisfy as N grows, the expected profit $\mathbb{E}[G_N]$ first increases then does not increase, so a unique optimal ticket count N^* exists. It is characterised by

$$N^* = 1 + \max\{N \in \mathbb{N}_0 : \mathbb{P}(S_N < S) > \frac{b}{a+b}\},$$

i.e. N^* is the last ticket whose sale is (strictly) profitable according to part a).

Step 3: Finding N^* numerically. With $a = 300$ and $b = 500$,

$$\frac{b}{a+b} = \frac{500}{800} = \frac{5}{8} = 0.625.$$

Using the CLT approximation, the profitability condition for selling the $(N + 1)$ -th ticket is

$$\Phi\left(\frac{S - \frac{1}{2} - Np}{\sqrt{Npq}}\right) > 0.625.$$

Since Φ is strictly increasing, this is equivalent to

$$\frac{S - \frac{1}{2} - Np}{\sqrt{Npq}} > z^* := \Phi^{-1}(0.625) \approx 0.32,$$

Therefore, the profitability criterion for selling the $(N + 1)$ -th ticket is fulfilled if and only if

$$\frac{S - \frac{1}{2} - Np}{\sqrt{Npq}} > 0.32.$$

Step 4: Evaluating the condition. We now evaluate this profitability criterion for the marginal changes around the capacity boundaries of both aircraft.

- **Airbus 320** ($S = 180$).

Check $N = 187$, i.e., the profit of the 188th ticket relative to the 187th:

$$\frac{179.5 - 0.95 \cdot 187}{\sqrt{0.0475 \cdot 187}} = \frac{179.5 - 177.65}{\sqrt{8.8825}} = \frac{1.85}{2.980} \approx 0.621 > 0.32.$$

Thus, expanding ticket sales from 187 to 188 is profitable.

Check $N = 188$:

$$\frac{179.5 - 0.95 \cdot 188}{\sqrt{0.0475 \cdot 188}} = \frac{179.5 - 178.60}{\sqrt{8.930}} = \frac{0.90}{2.988} \approx 0.301 < 0.32.$$

Thus, selling the 189th ticket is not profitable. Since the profitability condition is satisfied for $N = 187$ but fails for $N = 188$, the optimal number of tickets for the Airbus 320 is:

$$N^* = 188.$$

- **Boeing 747** ($S = 383$).

Check $N = 401$:

$$\frac{382.5 - 0.95 \cdot 401}{\sqrt{0.0475 \cdot 401}} = \frac{382.5 - 380.95}{\sqrt{19.048}} = \frac{1.55}{4.364} \approx 0.355 > 0.32.$$

Hence, selling the 402nd ticket is profitable.

Check $N = 402$:

$$\frac{382.5 - 0.95 \cdot 402}{\sqrt{0.0475 \cdot 402}} = \frac{382.5 - 381.90}{\sqrt{19.095}} = \frac{0.60}{4.370} \approx 0.137 < 0.32.$$

Hence, selling the 403rd ticket is not profitable. Since the profitability condition is satisfied for $N = 401$ but fails for $N = 402$, the optimal number of tickets for the Boeing 747 is:

$$N^* = 402.$$

□